

Creating Accessible Presentations With ltx-talk

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- ```
\DocumentMetadata{
 lang=en,
 tagging=on,
 pdfstandard=ua-2,
% check-tagging-status,
 tagging-setup={
% math/alt/use, % <=> Formulas must have description/alt text
% role/new-tag=frametitle/H1, % <=> headings must begin at level 1
 math/setup=mathml-SE
 }
}
\documentclass{ltx-talk}
```
- Lua $\text{\LaTeX}$
- tagged PDF + overlays: `\command<xyz>`  $\rightarrow$  `\command` only on slides xyz
- defined colors: **alert**, **example**, **structure**
- `{frame*}`  $\Leftrightarrow$  `\verb|...|`, `{verb}`, `{lstlisting}`, etc.

- Overlays: `\command<xyz>` → `\command` only on slides `xyz`
- Color commands with overlays: `\color`, `\mathcolor`, `\textcolor`; `\alert`
- Text commands with overlays: `\textbf`, `\textit`, `\textmd`, `\textnormal`, `\textrm`, `\textsc`, `\textsf`, `\textsl`, `\texttt`, `\textup`, `\emph`
- Also: `\includegraphics`; `\label`
- Also: `\item` (and can take an *action spec*)
- Main overlay commands: `\onslide`, `\only`, `\uncover`, `\visible`, `\invisible`, `\alt`, `\temporal`
- Overlay environments: `{onlyenv}`, `{invisibleenv}`, `{uncoverenv}`, `{visibleenv}`

## The Basics

## Sectioning

## Examples

- List Environments

- Interesting Use of Overlays

## Customizations

- Adjusting the Header and Footer

- Adjusting the Title Page

- Creating a Custom Title and Footer Element

- More Customizations

## Future Work

---

Default tocdepth is 2 (\subsection).

## The Basics

## Sectioning

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Adjusting the Header and Footer

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---

Default `tocdepth` is 2 (`\subsection`).

- `\part`, `\chapter`,  
`\section`, `\subsection`, `\subsubsection`,  
`\paragraph`, `\subparagraph`
- (currently?) no visible output  
only affects the `\tableofcontents`

- `{itemize}`, `{enumerate}`, `{description}` as in usual classes
- templates mean we can pass options to the environments, see `texdoc blocks-doc`
- extra (default) key `action-spec`
- `action-spec=<+->` uncover one item at a time
- `action-spec=<+-|alert@+>` and `\alert` the new item

```
\newlength\mylabelwidth
\setlength\mylabelwidth
 {\widthof{tagged PDFs }}\
\begin{frame}
\frametitle{A Strangely Sporadic Glossary}
\begin{description}[
 label-width=\mylabelwidth,
 label-align=left,
 left-margin=\mylabelwidth,
 action-spec=<+>,]
\item[\LaTeX] \TeX\ and one or two
 additional macros
\item[\texttt{beamer}] presentation
 class for \LaTeX
\item[tagged PDFs] Computers can read PDFs
\item[\texttt{ltx-talk}] tagged
 \texttt{beamer} PDF
\end{description}
\end{frame}
```

**LaTeX**

TeX and one or two additional macros

**beamer**

presentation class for LaTeX

**tagged PDFs**

Computers can read PDFs

**ltx-talk**

tagged beamer PDF



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\end{description}
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TeX and one or two additional macros

**beamer**

presentation class for LaTeX

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\end{description}
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\end{description}
\end{frame}
```

**$\LaTeX$**

$\TeX$  and one or two additional macros

**beamer**

presentation class for  $\LaTeX$

**tagged PDFs**

Computers can read PDFs

**ltx-talk**

tagged beamer PDF

- Just like in Beamer
- To define an overlay command:

```
\NewDocumentCommand{\mycommand}{ s D<>{1-} ... }{ ...}
% or
\NewDocumentCommand{\mycommand}{ s d<> ... }{ ...}
```

- For example

```
\NewDocumentCommand{\visablealert}{ d<> m }{%
 \visible<#1->{\alert<#1>{#2}}}
```

```
\begin{frame}[tag-slides={1-}]
\frametitle{Matrix Multiplication Example,
Step \arabic{slide}}
\NewDocumentCommand{\visablealert}{d<>m}{%
\visible<#1->{\alert<#1>{#2}}}
\newcommand{\colvect}[1]{%
\begin{pmatrix}#1\end{pmatrix}}
\begin{gather*}
\begin{pmatrix}
\alert<2,3>{1}&\alert<2,3>{2}\\
\alert<4,5>{4}&\alert<4,5>{5}
\end{pmatrix}
\begin{pmatrix}
\alert<2,4>{-2}&\alert<3,5>{4}\\
\alert<2,4>{2}&\alert<3,5>{-5}
\end{pmatrix}
\visable<2-6>{
{}}=\begin{pmatrix}
```

$$\begin{pmatrix} 1 & 2 \\ 4 & 5 \end{pmatrix} \begin{pmatrix} -2 & 4 \\ 2 & -5 \end{pmatrix}$$

```
\NewDocumentCommand{\visablealert}{d<>m}{%
 \visible<#1->{\alert<#1>{#2}}}
\newcommand{\colvect}[1]{%
 \begin{pmatrix}#1\end{pmatrix}
\begin{gather*}
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\end{pmatrix}
\begin{pmatrix}
\alert<2,4>{-2}&\alert<3,5>{4}\\
\alert<2,4>{2} &\alert<3,5>{-5}
\end{pmatrix}
\end{gather*}
\visible<2-6>{
 { }=\begin{pmatrix}
 \visablealert<2>{2}&
 \visablealert<3>{-6}\\
 \visablealert<4>{2}&
```

$$\begin{pmatrix} 1 & 2 \\ 4 & 5 \end{pmatrix} \begin{pmatrix} -2 & 4 \\ 2 & -5 \end{pmatrix} = \begin{pmatrix} 2 & \end{pmatrix}$$

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 2 \end{pmatrix} = 2$$

```
\begin{pmatrix}
\alert<2,3>{1}&\alert<2,3>{2}\\
\alert<4,5>{4}&\alert<4,5>{5}
\end{pmatrix}
\begin{pmatrix}
\alert<2,4>{-2}&\alert<3,5>{4}\\
\alert<2,4>{2} &\alert<3,5>{-5}
\end{pmatrix}
\visible<2-6>{
 {}=\begin{pmatrix}
 \visablealert<2>{2}&
 \visablealert<3>{-6}\\
 \visablealert<4>{2}&
 \visablealert<5>{-9}
 \end{pmatrix}}
\\[\bigskipamount]
\phantom{\colvect{1}\2}}%
\visible<2-5>{
```

$$\begin{pmatrix} 1 & 2 \\ 4 & 5 \end{pmatrix} \begin{pmatrix} -2 & 4 \\ 2 & -5 \end{pmatrix} = \begin{pmatrix} 2 & -6 \end{pmatrix}$$

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ -5 \end{pmatrix} = -6$$

```
\begin{pmatrix}
\alert<2,4>{-2}&\alert<3,5>{4}\\
\alert<2,4>{2} &\alert<3,5>{-5}
\end{pmatrix}
\visible<2-6>{
{}=\begin{pmatrix}
\visiblealert<2>{2}&
\visiblealert<3>{-6}\\
\visiblealert<4>{2}&
\visiblealert<5>{-9}
\end{pmatrix}}
\\[\bigskipamount]
\vphantom{\colvect{1}\2}}%
\visible<2-5>{
\alert{
\only<2,3>{\colvect{1}\2}}
\only<4,5>{\colvect{4}\5}}
\bullet
```

$$\begin{pmatrix} 1 & 2 \\ 4 & 5 \end{pmatrix} \begin{pmatrix} -2 & 4 \\ 2 & -5 \end{pmatrix} = \begin{pmatrix} 2 & -6 \\ 2 & -9 \end{pmatrix}$$

$$\begin{pmatrix} 4 \\ 5 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 2 \end{pmatrix} = 2$$



```
\visible<2-6>{
 {}=\begin{pmatrix}
 \visablealert<2>{2}&
 \visablealert<3>{-6}\\
 \visablealert<4>{2}&
 \visablealert<5>{-9}
 \end{pmatrix}}
\\[\bigskipamount]
\vpantom{\colvect{1}\2}}%
\visible<2-5>{
 \alert{
 \only<2,3>{\colvect{1}\2}}
 \only<4,5>{\colvect{4}\5}}
 \bullet
 \only<2,4>{\colvect{-2}\2}}
 \only<3,5>{\colvect{4}\-5}}
 =
 \only<2>{2}\only<3>{-6}
```

$$\begin{pmatrix} 1 & 2 \\ 4 & 5 \end{pmatrix} \begin{pmatrix} -2 & 4 \\ 2 & -5 \end{pmatrix} = \begin{pmatrix} 2 & -6 \\ 2 & -9 \end{pmatrix}$$

$$\begin{pmatrix} 4 \\ 5 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ -5 \end{pmatrix} = -9$$

```
\visiblealert<3>{-6}\\
\visiblealert<4>{2}&
\visiblealert<5>{-9}
\end{pmatrix}}
\\[\bigskipamount]
\vphantom{\colvect{1}\2}}%
\visible<2-5>{
\alert{
\only<2,3>{\colvect{1}\2}}
\only<4,5>{\colvect{4}\5}}
\bullet
\only<2,4>{\colvect{-2}\2}}
\only<3,5>{\colvect{4}\-5}}
=
\only<2>{2}\only<3>{-6}
\only<4>{2}\only<5>{-9}}}}
\end{gather*}
\end{frame}
```

$$\begin{pmatrix} 1 & 2 \\ 4 & 5 \end{pmatrix} \begin{pmatrix} -2 & 4 \\ 2 & -5 \end{pmatrix} = \begin{pmatrix} 2 & -6 \\ 2 & -9 \end{pmatrix}$$

- Which slides should a screen reader use?
- Default is the last slide: `tag-slides=n`
- Previous example: `tag-slides={1-}`
- Which slides should `\documentclass[mode=handout]{ltx-talk}` use?
- Default is to ignore all overlays
- Fails in previous example:

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} \begin{pmatrix} 4 \\ 5 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 2 \end{pmatrix} \begin{pmatrix} 4 \\ -5 \end{pmatrix} = 2 - 62 - 9$$

`\EditInstance{header}{std}{<key=value>}` or  
`\EditInstance{frametitle}{header}{font=<value>}`

| Key               | Default Value                         |
|-------------------|---------------------------------------|
| background-color  | (none)                                |
| color             | structure                             |
| font              | \normalfont                           |
| height            | topmargin (10 mm)<br>+ headsep (2 mm) |
| left-hspace       | leftmargin (10 mm)                    |
| right-hspace      | rightmargin (10 mm)                   |
| print-frame-title | true                                  |



```
\NewCommandCopy\oldframetitle\frametitle
\renewcommand{\frametitle}[1]{%
 \oldframetitle{#1\hfill
 \includegraphics[height=2ex,artifact]{logo-file}%
 }%
}
```

```
\EditInstance{footer}{std}{<key=value>}
```

| Key              | Default Value       |
|------------------|---------------------|
| background-color | (none)              |
| color            | black               |
| element-order    | (empty)             |
| font             | \tiny               |
| left-hspace      | leftmargin (10 mm)  |
| right-hspace     | rightmargin (10 mm) |
| separator        | \hfil               |

Available footer elements:

author, date, institute, title, subtitle, framenumbers, totalframes

(=`\RefProperty{lastpage}{totalframes}`)

For example:

```
\EditInstance{footer}{std}{element-order={date,author,title,framenumbers}}
```

```
\maketitle[
 element-order = {
 title, subtitle, author, institute, date
 },
 framestyle = talk,
 horizontal-alignment = center,
 vertical-alignment = center
]
```

framestyle = pagestyle  
other options are plain & wallpaper

```
\DeclareInstance{titlepage-element}{myelement}{talk}{<key=value>}
% and then
\makeatletter
\newcommand{@myelement}{...}
\makeatother
% or
\ExpandArgs{c}\newcommand{@myelement}{...}
```



```
\DeclareInstance{footer-element}{myelement}{talk}{...}
% and then
\makeatletter
\newcommand\@myelement{...}
% or
\newcommand\@shortmyelement{...}
\makeatother
% or
\ExpandArgs{c}\newcommand{\@myelement}{...}
% or
\ExpandArgs{c}\newcommand{\@shortmyelement}{...}
```

(\@shortmyelement will be preferred if available)

```
\EditInstance{floatenv}{std}{horizontal-alignment = <alignment>}
% left (default), center, or right
```

```
\EditInstance{hidden}{std}{opacity = <value>}
% 0 <= value <= 1 for how opaque hidden things should be
```

- better tagging
- better heading structure?